

SRIKANTH YELEM

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Summary

Electrical Engineer with 4+ years of experience delivering power system, substation, and building electrical solutions across consulting, utility, and EPC environments. Applied load flow, short-circuit, arc flash, and protection coordination studies to improve system reliability, fault selectivity, and safety performance. Led renewable integration, SCADA-based monitoring enhancements, and substation design initiatives, strengthening grid visibility and operational efficiency. Strong expertise in IEC/IEEE standards, grounding design, and equipment engineering, ensuring compliant, resilient, and maintainable electrical infrastructure across project lifecycle.

Skills

Power Systems Engineering: Load Flow Analysis, Short Circuit Studies, Stability Analysis, Power System Modeling, Contingency Analysis, Power Quality Analysis, Grid Operations & Monitoring, Renewable Energy Integration (Solar), Grid Interconnection Studies

Transmission & Substation Engineering: Substation Design (AIS/GIS), Single Line Diagrams (SLD), Substation Layout Engineering, Transformer & Switchgear Selection, CT/PT Sizing & Application, Cable Sizing & Routing, Earthing & Grounding Design (IEEE 80), Insulation Coordination

Protection, Relaying & Substation Automation: Distance, Differential, Overcurrent & Earth Fault Protection, Protection Coordination Studies, Relay Setting Calculations, Numerical Relays (Siemens, ABB, Schneider Electric), IEC 61850 Protocol, SCADA Systems, Fault Analysis & Troubleshooting

Renewable Integration & Grid Modernization: Inverter-Based Resource Modeling, Distributed Energy Resource (DER) Interconnection Studies, Grid-Forming & Grid-Following Control Strategies, Battery Energy Storage Systems (BESS), Microgrid Planning & Design, Advanced Distribution Management Systems (ADMS)

Standards & Regulatory Compliance: IEEE & IEC Standards, IS Standards, CEA Regulations, Indian Electricity Grid Code (IEGC), Grid Compliance & Safety Standards

Engineering Tools & Software: ETAP, PSS®E, OpenDSS, PowerWorld, ASPEN OneLiner, MATLAB/Simulink, PSCAD (Power System Transients), Typhoon HIL Simulator (Real-Time Simulations), AutoCAD Electrical, Advanced Microsoft Excel

Professional Experience

Electrical Engineer-Internship, *Delta Electronics*

June 2026 – Present | NC, USA

- Performing system modeling and studies for AC/DC Microgrids for datacenter applications.
- Performed load flow and voltage regulation studies using ETAP and PSS®E for a 12 MW mixed-use facility, evaluating system performance under varying load conditions and validating design adequacy.
- Developed short-circuit and arc flash studies using PSCAD for multiple distribution panels, establishing protection coordination and reducing incident energy exposure across critical electrical infrastructure systems.
- Executed grid interconnection and backup generation studies using PSCAD for 8 MW standby systems, assessing DER interactions, fault contributions, and ensuring compliance with utility and interconnection standards.
- Applied contingency analysis and power system modeling to evaluate system performance during outage scenarios, incorporating inverter-based resource behavior to enhance resilience of healthcare electrical systems under critical operating conditions.

Electrical Engineer, *VRVS India Pvt. Ltd*

Jan 2016 – Dec 2017 | AP, India

- Performed load flow, short-circuit, and stability studies using PSS®E and PSCAD for a 132 kV transmission network expansion, improving voltage stability and ensuring reliable grid operation under varying demand conditions.
- Conducted DER interconnection studies using OpenDSS for 60 MW solar integration at distribution level, evaluating voltage fluctuations, fault levels, and ensuring compliance with grid codes for stable network performance.
- Developed inverter-based resource models using MATLAB/Simulink, analyzing grid-forming and grid-following control strategies to support renewable integration and improve system stability under dynamic and transient grid operating conditions.

- Analyzed power quality issues across multiple industrial feeders, identifying harmonic distortions and voltage deviations, implementing corrective measures to enhance system performance and reduce equipment stress within transmission network infrastructure.
- Executed contingency analysis and fault simulations using PowerWorld and PSCAD, identifying overload risks and supporting corrective strategies to maintain system reliability during peak demand and unexpected transmission outages.
- Enhanced SCADA-based monitoring across transmission assets, improving real-time fault detection, operational visibility, and coordination efficiency for grid operators managing large-scale generation and transmission infrastructure systems.
- Ensured compliance with IEGC, IEEE, and IEC standards while supporting grid planning and operational studies, maintaining system reliability and facilitating regulatory approvals for transmission and power generation projects.

Electrical Engineer, Futuronix

Jun 2012 – Aug 2013 | AP, India

- Designed 132 kV AIS substations including single line diagrams and layouts using AutoCAD Electrical, ensuring compliance with specifications and optimizing space utilization for installation, operation, and long-term maintenance accessibility.
- Performed transformer and switchgear selection for high-capacity loads, applying insulation coordination principles to ensure reliable equipment performance across transmission and industrial electrical infrastructure projects.
- Engineered earthing and grounding systems using IEEE 80 standards, ensuring safe step and touch potentials while enhancing personnel safety under high fault current conditions within substation environments.
- Conducted protection coordination studies and relay setting calculations using ETAP and ASPEN OneLiner, ensuring selective fault isolation and reliable operation of overcurrent and earth fault protection schemes.
- Supported IEC 61850-based substation automation implementation, enabling efficient communication, improved monitoring, and enhanced control capabilities across electrical systems within large-scale infrastructure and EPC project environments.
- Reviewed vendor technical specifications and engineering drawings, ensuring alignment with design requirements, minimizing discrepancies, and supporting efficient procurement and execution phases across substation and transmission infrastructure projects.

Education

Ph.D, University of Missouri

Jan 2020 – Present | MO, USA

Electrical and Computer Engineering

Master of Science, University of Missouri

Jan 2018 – Dec 2019 | MO, USA

Electrical Engineering